

Python Cheat Sheet: ML, AI & GenAI Essentials

Lists

Ordered, mutable, allows duplicates.

Syntax:

```
my_list = [1, 2, 3, 'AI']
```

```
my_list.append(4)
```

```
my_list[0] # 1
```

Use: Predictions, prompts

Dictionaries

Unordered, key-value pairs, mutable, keys unique.

Syntax:

```
my_dict = {'model': 'GPT-4', 'accuracy': 0.95}
```

```
my_dict['model'] # 'GPT-4'
```

Use: Model scores, API payloads

Tuples

Ordered, immutable, allows duplicates.

Syntax:

```
my_tuple = (28, 28)
```

Use: Input shapes, multiple return values

Sets

Unordered, unique values, mutable.

Syntax:

```
my_set = {1, 2, 2, 3} # {1, 2, 3}
```

Use: Unique labels, deduplication

Comprehensions

Efficient syntax to build collections.

List: `[x**2 for x in range(5)]`

Dict: `{x: x**2 for x in range(3)}`

Set: `{x for x in [1, 2, 2, 3]}`

Use: Data transformation, prompts

Functions

Reusable blocks of logic.

Syntax:

```
def greet(name):
```

```
    return f"Hello, {name}"
```

Use: Metrics, prompts, reusable logic

map()

Applies function to each item.

Syntax:

```
list(map(lambda x: x.upper(), ['ai', 'ml']))
```

Use: Format outputs, batch processing

filter()

Filters elements based on condition.

Syntax:

```
list(filter(lambda x: x > 0.5, [0.2, 0.9, 0.6]))
```

Use: Filter predictions, valid outputs

reduce()

Collapse sequence to a single value.

Syntax:

```
from functools import reduce  
reduce(lambda a, b: a + b, [1, 2, 3]) # 6
```

Use: Total score, build sentence

Quick Comparison Table

Concept	Ordered	Mutable	Duplicates	Use Case
List	Yes	Yes	Yes	Predictions, prompts
Dictionary	No	Yes	No (keys)	Model scores, configs
Tuple	Yes	No	Yes	Input shapes, returns
Set	No	Yes	No	Deduplication, unique values
Comprehension	-	-	-	Efficient generation
Function	-	Yes	-	Reusable logic
map()	Yes	-	-	Transformation
filter()	Yes	-	-	Selection